

Session7: Intro to Information Security..



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7.0 Vulnerability, Threat and Risk

Outcome:

1. You'll be able to understand the definition of vulnerability in security domain.
2. You'll be able to understands the types of vulnerabilities.
3. . You'll be able to have deep understanding the difference between Threat and the Risk in Security.



7.1 Vulnerability

- It is the weakness in a mechanism that can threaten the wellbeing of an asset.
- In other words, it is the lack of Countermeasures of a mechanism or a system.
- Real life Example: When you are roaming around without a mask, you are vulnerable to any disease including covid-19.
- IT world example: Running a system without a firewall, your system is vulnerable to any attack.



7.1.1 Types of Vulnerabilities

- **Design Vulnerability** - When the Vulnerability is said to be inherent to the project or design.
- **Implementation Vulnerability** - referred to as low-level flaws or technical flaws.
- **Configurational Vulnerability** - A flaw in your security settings, like failing to auto-encrypt your files, could leave your entire network and every device connected to it vulnerable to any attack.



7.2 Threat

- Threat is any potential danger to your information or system.
- Someone uncovering a vulnerability and exploiting it. That person becomes the threat to your organization.

7.3 Risk



- Business impact and the probability of a vulnerability being exploited.
- For example if you are not carrying a backup regularly in your organization, you are at risk.
- In the next section, we will discuss on how we manage risks.



7.3.1 Risk Management

- You as the manager for the data center of your organization, one of your prime roles is to
 - (a) identify the risks,
 - (b) assess and prioritize them,
 - (c) control the risks and
 - (d) review the controls.



Questions???

- Any Questions ???